



NAM HOUNG - 1 HYDRO POWER PROJECT



Project title	Nam Houg – 1 Hydro Power Project
Project ID	4588
Monitoring period	01-September-2023 to 31-December-2024 (Inclusive of both days)
Original date of issue	11-March-2025
Most recent date of issue	14-May-2025
Version	02
VCS Standard Version	4.7
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PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

Nam Houng - 1 Hydro Power Project (hereafter referred to as the “the project”) is developed by Nam Houng - 1 Hydro Power Co., Ltd in Xayabouly province, Lao PDR. The primary purpose of the project is to generate electricity to the Lao Power Grid.

The project activity is a hydro power project with a total installed capacity of 15 MW. The project is promoted by Nam Houng - 1 Hydro Power Co., Ltd. Project was commissioned on 01-September-2023 with a net annual generation of 87,280 MWh.

The project is expected to constantly contribute clean energy to the Lao Power Grid. It not only conforms to the present situation, but also can adapt to future changes in the electricity market. The baseline scenario of the project is continuation of the present situation, i.e. electricity supplied from the power grid. By displacing part of the power generated by thermal power plants, the project is therefore expected to reduce CO₂ emissions by an estimation of 31,453 tCO_{2e} per year during the first crediting period of 7 years.

Nam Houng - 1 Hydro Power Co., Ltd. is a SPV of Simoung Group Co., Ltd. Simoung Group Co., Ltd. is the legal owner of Nam Houng – 1 Hydro Power Project (Nam Houng - 1 Hydro Power Co., Ltd.). They have the legal right to control and operate the project activities.

In the current monitoring period from 1-September-2023 to 31-December-2024 (including both the dates) the project activity has supplied 72,978 MWh of electricity and thus contributing to total CO₂ emission reductions of 26,298 tCO_{2e}.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	13-December-2023 – 29-February-2023	VCS	4K Earth Science Private Limited	-
Verification	01-September-2023 – 31-December-2024	VCS	4K Earth Science Private Limited	1 year 04 months

1.3 Sectoral Scope and Project Type

Sectoral scope ¹	01-Energy Industries (Renewable/non-renewable sources)
Project activity type	Renewable Energy Project

The project is not AFOLU project and is not a grouped project.

1.4 Project Proponent

Organization name	Nam Houg – 1 Hydro Power Co., Ltd
Contact person	Phapithack Ekaphanh
Title	Director
Address	Khampheng Mueang Road (T4), Phonthan Nuea Village, Sayysathar District, Vientiane Capital
Telephone	+85621 922777
Email	smgroad@yahoo.com infor@smgroup.com.la

1.5 Other Entities Involved in the Project

Organization name	Kosher Climate India Private Limited
Role in the project	Project Representative
Contact person	Narendra Kumar Ramaraj
Title	Director- Technical
Address	Zee Plaza, No.1678, 27th Main Rd, Near KLM Mall, Sector 2, HSR Layout, Bengaluru, Karnataka 560102, India
Telephone	+919632803444
Email	narendra@kosherclimate.com

1.6 Project Start Date

Project start date	01-September-2023
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¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

Justification	Commercial Operation Date
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1.7 Project Crediting Period

Crediting period	☒ <i>Seven years, twice renewable</i> ☐ <i>Ten years, fixed</i> ☐ <i>Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)</i>
Start and end date of first or fixed crediting period	01-September-2023 to 31-August-2030 (including both dates)

1.8 Project Location

The project is located in the Xayabouly Province, Lao PDR.

Project Location	Latitude	Longitude
Xayabouly Province, Lao PDR	19° 11'19.7"N	101° 48'17.8"E

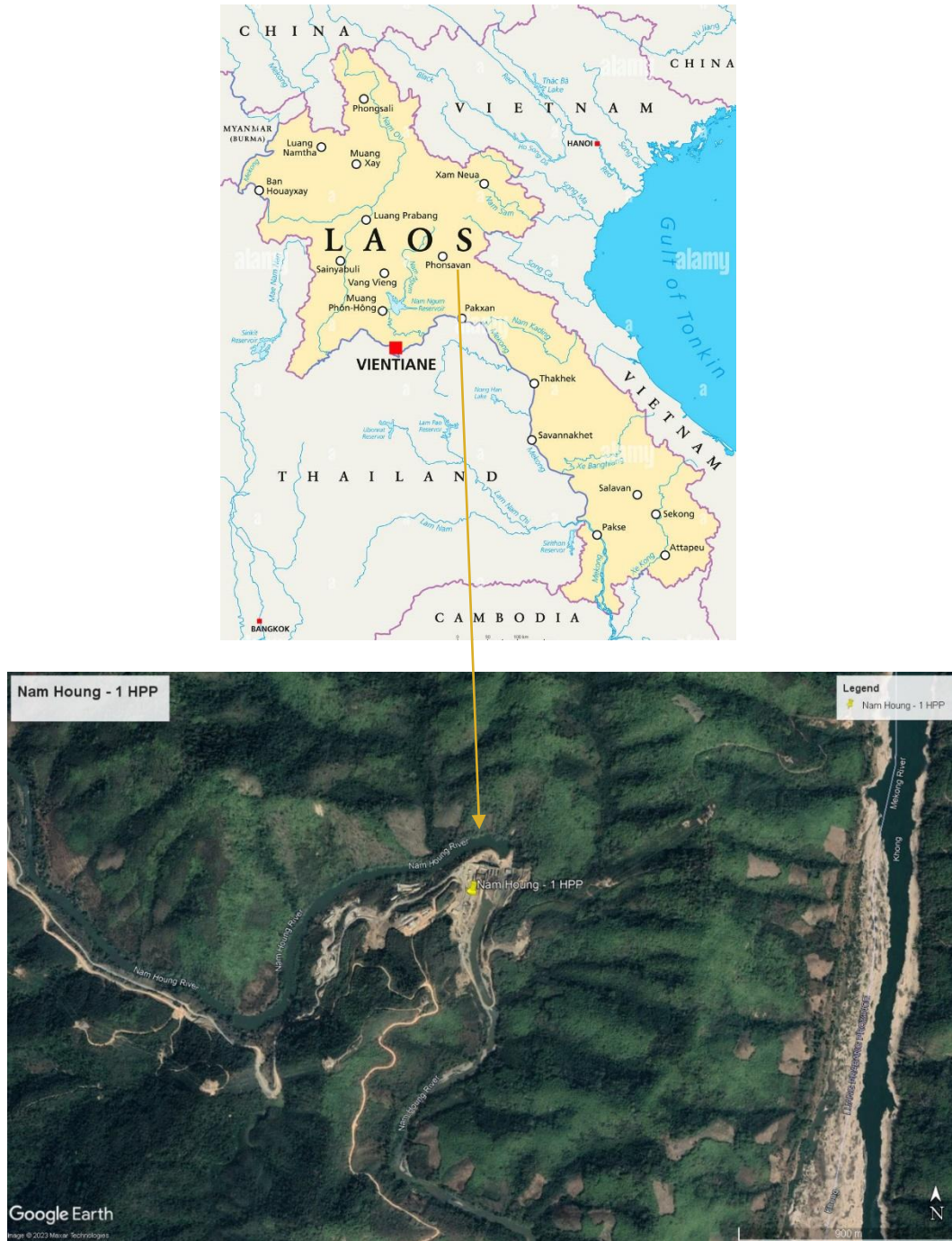


Fig.1 Project Location

1.9 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS I. D ²	Small-scale methodology: Grid-connected renewable electricity generation	18.0
Methodology ³	ACM0002 ⁴	Grid-connected electricity generation from renewable sources	21.0
Tool	CDM TOOL07 ⁵	Tool to calculate the emission factor for an electricity system	07.0
Tool	CDM TOOL21 ⁶	Demonstration of additionality of small-scale project activities	13.1
Tool	CDM TOOL27 ⁷	Investment analysis	12.0

1.10 Double Counting and Participation under Other GHG Programs

1.10.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.10.2 Registration in Other GHG Programs

Was the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

² <https://cdm.unfccc.int/UserManagement/FileStorage/2P7FS6ZQAR84LG3NMKYUH50WI9ODBC>

³ Methodology referred for calculation of project emissions as per AMS -I.D

⁴ <https://cdm.unfccc.int/UserManagement/FileStorage/ZPFJL01OU2RYC6N3HASIXV7K84QBG9>

⁵ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

⁶ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-21-v13.1.pdf>

⁷ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf>

1.11 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.11.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit?

☐ Yes ☒ No

1.11.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system?

☐ Yes ☒ No

1.11.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.12 Sustainable Development Contributions

The project has been contributed to 03 SDGs which are 7, 8, and 13.

The project's quantifiable contributions to specific targets and indicators of the Sustainable Development Goals (SDGs) for the SD contributions reporting period is detailed in following table.

The project has been contributed to sustainable development using the following ways.

- **Social well-being:** The project would help in generating employment opportunities during the construction and operation phases. The project activity leads to development in infrastructure in the region like development of roads and also may promote business with improved power generation.
- **Economic well-being:** The project is a clean technology investment in the region, which would not have been taken place in the absence of the VCS benefits the project activity will also help to reduce the demand supply gap in the state. The project activity will generate power using hydro energy-based power generation which helps to reduce GHG emissions and specific pollutants like SO_x, NO_x, and SPM associated with the conventional thermal power generation facilities.
- **Technological well-being:** The successful operation of project activity would lead to promotion of hydro power generation and would encourage other entrepreneurs to participate in similar projects.

- Environmental well-being: Hydro energy being a renewable source of energy, it reduces the dependence on fossil fuels and conserves natural resources which are on the verge of depletion. Due to its zero emission the Project activity also helps in avoiding significant amount of GHG emissions.

As per para 3.17.1 of the VCS standard (v4.7), the requirement prescribes that “The project proponent shall demonstrate how the project activities, or additional activities implemented by the project proponent, contribute to sustainable development, as defined by and tracked against the United Nations Sustainable Development Goals (SDGs). The project proponent must demonstrate that a project contributes to at least three SDGs by the end of the first monitoring period and in each subsequent monitoring period”.

In continuation, the project has achieved SDGs during the current monitoring period which fulfils the requirement of minimum three SDG indicators. These contributions are separately reported in VERRA prescribed SDG Report and submitted to VERRA. Also, during the verification process supporting documents related to such SDG contributions were submitted to VVB; the summarized details of the contributions are provided under table1 below: -

Table 1: Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime								
1)	7.2	7.2.1 Renewable energy share in the total final energy consumption	Implemented activities to increase	During the reported monitoring period (01-September-2023 to 31-December-2024), the net renewable energy supplied to grid is 72,978 MWh.	The project supplied 72,978 MWh renewable electricity to grid since the COD till end of the monitoring period. <table><tr><th>Monitoring Period</th><th>Electricity Generation (MWh)</th></tr><tr><td>1-September-2023 to 31-December-2023</td><td>24,107</td></tr><tr><td>1-January-2024 to 31-December-2024</td><td>48,871</td></tr><tr><td>Total</td><td>72,978</td></tr></table>	Monitoring Period	Electricity Generation (MWh)	1-September-2023 to 31-December-2023	24,107	1-January-2024 to 31-December-2024	48,871	Total	72,978
Monitoring Period	Electricity Generation (MWh)												
1-September-2023 to 31-December-2023	24,107												
1-January-2024 to 31-December-2024	48,871												
Total	72,978												
2)	8.5	8.5.1 Average hourly earnings of female and male employees, by occupation, age and persons with disabilities.	Implemented activities to increase	Total of 17 employment opportunities were provided by the project owner related to the project activity	The project generated 17 new employment opportunities since the COD								

3)	13.2	13.2.2 Total greenhouse gas emissions per year	Implemented activities to increase	By supplying 72,978 MWh of clean electricity to the grid, the project avoided release of 26,298 tCO ₂ into the atmosphere during the reporting monitoring period (01-September-2023 to 31-December-2024) and contributed to the sustainable development goal 13.	Prevented the release of 26,298 tCO₂ into the atmosphere since its commissioning till the end of the monitoring period.								
					<table><tr><th>Monitoring Period</th><th>Emission Reductions (tCO₂e)</th></tr><tr><td>1-September-2023 to 31-December-2023</td><td>8,687</td></tr><tr><td>1-January-2024 to 31-December-2024</td><td>17,611</td></tr><tr><td>Total</td><td>26,298</td></tr></table>	Monitoring Period	Emission Reductions (tCO ₂ e)	1-September-2023 to 31-December-2023	8,687	1-January-2024 to 31-December-2024	17,611	Total	26,298
Monitoring Period	Emission Reductions (tCO ₂ e)												
1-September-2023 to 31-December-2023	8,687												
1-January-2024 to 31-December-2024	17,611												
Total	26,298												

1.13 Commercially Sensitive Information

No commercial sensitive information has been excluded from the public version of the project description.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	There is no change involved in the stakeholder identification since validation. Therefore, this parameter is not applicable.
Legal or customary tenure/access rights	The project is built and owned by Nam Houg – 1 Hydro Power Co., Ltd. Nam Houg - 1 Hydro Power Co., Ltd. is a SPV of Simoung Group Co., Ltd. Simoung Group Co., Ltd. is the legal owner of Nam Houg – 1 Hydro Power Project (Nam Houg - 1 Hydro Power Co., Ltd.). They have the legal right to control and operate the project activities. The business license of the project owner are the evidences for legislative right. Besides, the project allotment license, the equipment purchase contract and the power purchase agreement are the evidences for the ownership of the plant equipment and power generation. There are no identified disputes or conflicts over rights of the land and its resources. No concerns were received against the implementation of the project activity and confirms the consent and full support for the project implementation.

Stakeholder diversity and changes over time	Each community has its own cultural practices, customs, and beliefs, contributing to the overall cultural heritage and integrity of the region. The project anticipates no significant changes to the group's cultural integrity in the project region. Further, the project activities by contributing to the environmental, economic, and social benefits, play a crucial role in enabling local communities to remain located in the project area.
Expected changes in well-being	Environmental well-being: Hydro power is one of the cleanest renewable energy powers and does not involve any fossil fuel. There are no GHG emissions. The impact on land, water, air and soil is negligible. Thus, the project activity contributes to environmental well-being without causing any negative impact on the surrounding environment. Economic well-being: The electricity is being supplied to the grid as per PPA which indirectly helps in the increase of electricity in the country to enhance the economy and life style of the area. Social well-being: The project activity generated employment to the local people during the construction and operation of the project activity.
Location of stakeholders	All community members, those who are directly or indirectly impacted by the project are mostly local people who reside near the project areas.
Location of resources	The location of territories and resources to which stakeholders have customary access is the nearby land in Xayabouly Province, Lao PDR.

2.1.2 Stakeholder Consultation and Ongoing Communication

Ongoing consultation	A stakeholder engagement mechanism has been established to address concerns and resolve potential conflicts regarding the project between the project proponent and local stakeholders. Before the initial submission of PD, the project proponent conducted the LSC meeting about the project. There is a mechanism in place to engage the stakeholders, to address their grievances and
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	resolve any potential conflict between the project proponent and local stakeholders. The PP provided contact information, including email addresses and phone numbers, to stakeholders at the LSC meeting, enabling them to share ideas and express grievances about the project in the future. This is the ongoing consultation which has been followed for this monitoring period. Additionally, a grievance register has been set up to record comments from stakeholders. The project proponent is maintaining the grievance register at the project site and during this monitoring period, PP didn't receive any grievances.
Date(s) of stakeholder consultation	21-August-2022
Communication of monitored results	During the LSC meeting conducted before the initial PD submission, the Project Proponent (PP) provided a dedicated email address as well as the phone number and physical address of PP to all stakeholders to facilitating open and accessible communication channels for future engagement. A grievance register is being maintained at the site to register the complaints of the stakeholders. There was an active presence of the Project Proponent team in the project area. They have been worked closely with the community and other stakeholders to address any queries or grievances in relation to the project. Also assured that a grievance register was always made available at the project site to register their complaints if any in the register. For this monitoring period, there were no grievances registered.
Consultation records	A grievance register has been maintained at the site to register the complaints of the stakeholders.
Stakeholder input	No grievances/inputs were received from the stakeholders during the current monitoring period.

2.1.3 Free, Prior, and Informed Consent

Consent	There are no identified disputes or conflicts over rights of the land and its resources.
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	No concerns were received against the implementation of the project activity and confirms the consent and full support for the project implementation.
Outcome of FPIC	The project fully adhered to the Free, Prior, and Informed Consent (FPIC) process, ensuring that Indigenous Peoples (IPs), local communities (LCs), and customary rights holders were meaningfully engaged before project implementation. Stakeholder feedback was incorporated into project planning, ensuring alignment with community interests and rights. The project has not encroached on customary or communal lands, nor has it resulted in any involuntary physical or economic displacement. No individuals or communities were relocated without their free and informed consent, and no forced land acquisitions occurred.

2.1.4 Grievance Redress Procedure

Grievances received	Resolution and outcome
Not applicable	Project Proponent maintained the grievance register at the project site and made it accessible to all the stakeholders to raise their concerns or complaints regarding the project. During the current monitoring period, no grievances were received.

2.1.5 Public Comments

Summary of comments received	Actions taken
No concerns received during the public comment period and local stakeholder consultation. The potential benefits of the project activity for the local stakeholders were acknowledged. Local stakeholders were totally in support for setting up of these kinds of projects in the region.	There were no comments received.

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The project is managed by a highly experienced team with a strong track record in implementing similar projects and engaging local communities. The management team comprises experts in hydropower development, environmental management, and stakeholder engagement, with prior experience in successfully delivering sustainable energy projects. Their expertise includes compliance with international best practices, regulatory frameworks, and environmental and social safeguard policies.

Where specialized expertise was required, the project proponent has collaborated with technical consultants, environmental specialists, and community engagement organizations to ensure effective project implementation. Additionally, a structured recruitment strategy is in place to address any skill gaps by onboarding professionals with relevant experience in hydropower operations, ecological monitoring, and community relations. In cases where new entities have joined the project, their qualifications and experience in project design, environmental impact mitigation, and stakeholder engagement have been thoroughly assessed to ensure alignment with project goals. Through these measures, the project maintains high standards of execution, environmental stewardship, and inclusive community engagement.

2.2.2 Risk assessment

	Risk identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risks identified	Although no significant natural or human-induced risks to stakeholders' well-being have been identified, proactive mitigation measures will be implemented to ensure continued safety and environmental integrity. Seismic-resistant engineering designs and regular structural inspections will safeguard against unforeseen geological events. Flood management protocols, including real-time hydrological monitoring and controlled water discharge mechanisms, will prevent overflow risks. Occupational health and safety measures, such as worker training, PPE provisions, and emergency response plans, will ensure a risk-free working environment. Ongoing stakeholder engagement and grievance redress mechanisms will address any emerging concerns, ensuring that social

		harmony and project sustainability are maintained.
Risks to stakeholder participation	No risks identified	The project team has explained the project to the stakeholders including project benefits, costs, risks, rights, laws and opportunities on employment generation. The stakeholders are voluntarily willing to participate in the project because of benefits like safeguarding the communities from environmental challenges and various other SDG benefits of the project.
Working conditions	No risks identified	The project has taken comprehensive measures to ensure providing safe and healthy working conditions for its employees.
Safety of women and girls	No risks identified	The PP made sure for the safety of women and girls through the Company's Human Resource Management policy. The organization does not tolerate harassment for any women employees which includes sexual, psychological or any other form of harassment that can be a form of discrimination.
Safety of minority and marginalized groups, including children	No risks identified	The project poses no risks to the safety of minority and marginalized groups, including children, as there is no displacement, impact on vulnerable communities, or child labor involvement. Strict employment policies aligned with national and international labor laws prevent underage work, while controlled access to project sites ensures child safety. Social impact assessments and stakeholder consultations confirm no discrimination or exclusion risks. To maintain inclusivity and protection, the project will enforce access controls, conduct worker training on human rights, and implement a grievance redress

		mechanism, ensuring a safe and secure environment for all.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risks identified	Air emissions are minimal, as the project primarily relies on hydroelectric generation without combustion processes. Noise levels are controlled and localized to construction phases, with no long-term impact on surrounding communities. Water quality remains unaffected since the project does not involve industrial discharges, hazardous chemical use, or agricultural runoff. Solid and liquid waste generation is minimal, with proper disposal and recycling measures in place. Additionally, hazardous materials, including chemical pesticides and fertilizers, are not used in project activities, eliminating any contamination risks. Overall, the project complies with environmental standards and poses no pollution-related threats to stakeholders or ecosystems.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ⁸	Mitigation or preventative measure(s) taken
Discrimination	No risks identified	The organization has Human Resources Management Services policy in place to ensure there is no Discrimination and sexual harassment and would take appropriate steps for the prevention of such activities at workplace and put in place an appropriate mechanism of complaint/grievance redressal.

⁸ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

Sexual harassment	No risks identified	The organization has Human Resources Management Services policy in place to ensure there is no Discrimination and sexual harassment and would take appropriate steps for the prevention of such activities at workplace and put in place an appropriate mechanism of complaint/grievance redressal.
Gender equity in labor and work	No risks identified	The project abides the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. Specific to gender equality, detailed enforcement rules are regulated in Women's Rights Protection Law to provide support and benefits to women. Qualified local residents, both men and women, are recruited to work for the project. During stakeholders' consultation process, comments were collected from the local people, including both men and women. The project abides the rules of equality. Regarding the gender equality, detailed enforcement rules are regulated in Labor Law and Women's Rights Protection Law.
Forced labor	No risks identified	The project does not employ and is not complicit in any form of forced labor.
Child labor	No risks identified	The Project does not use victims of child labor. LAO PDRs "law on the protection of the Rights and Interests of children" prevents the employment of the minors. The Project requires skilled employees to operate, maintain, and manage the hydro power plant. Therefore, The project does not employ and is not complicit in any form of child labor. According to the Employee list, it could be

		confirmed that the Project does not employ any child labor.
Human trafficking	No risks identified	The Project does not use victims of human trafficking.

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risks identified	The Project Proponent emphasis on aligning the organizational practices with the human rights outlined in National and Regional Regulations. All customary rights to the land and its resources are held by Nam Houg-1 Hydro Power Co., Ltd. The project respects protection of the rights of local communities and customary rights holders, aligning with applicable International Human Rights Law, the United Nations Declaration.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risks identified	The project does not impact and protects any cultural heritage as part of project activities.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risks identified	The legal and customary rights of the land during and post implementation of the project activity lies with project proponent.

2.3.5 Benefit Sharing

Summary of the benefit sharing plan	Not applicable
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Benefit sharing during the monitoring period	Not applicable
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2.4 Ecosystem Health

	Risk identified	Mitigation or preventative measure(s) taken during the monitoring period
Impacts on biodiversity and ecosystems	No risk identified	With no identified risks to biodiversity and ecosystems, the project ensures continued protection through proactive monitoring and mitigation. An environmental management plan (EMP) and stakeholder engagement ensure adaptive management and early risk detection.
Soil degradation and soil erosion	No risk identified	There is no identified risks of soil degradation or erosion. Vegetative cover is maintained in exposed areas, while erosion control structures, such as silt traps and sediment barriers, prevent soil loss. Proper drainage systems manage water flow, reducing runoff impact.
Water consumption and stress	No risk identified	The project operates within natural flow regimes without significant diversion or depletion of local water sources. Optimized reservoir management prevents downstream shortages. Regular assessments and stakeholder engagement ensure water availability remains unaffected for local communities and ecosystems. No water stress is experienced in the location due to the project activity.

2.4.1 Rare, Threatened, and Endangered species

Species or habitat	The project is not located in or adjacent to habitats for rare, threatened, or endangered species. Hence, it is not applicable.
Areas needed for habitat connectivity	The project's design maintains natural water flow, preserving aquatic migration routes and ensuring unhindered wildlife movement. No critical corridors or

	breeding grounds have been disrupted, and surrounding vegetation remains intact. Preventive measures, such as buffer zones and erosion control, further support ecological continuity, ensuring long-term habitat connectivity.
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	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	No risk identified	A Comprehensive IEE report confirms that the project does not impact Habitats for rare, threatened, and endangered species. Baseline biodiversity studies and monitoring during the project period have found no presence of such species within or near the project area. The site selection process ensured avoidance of ecologically sensitive zones, and no critical habitats have been altered or fragmented.
Areas for habitat connectivity	No risk identified	Environmental assessments confirm that the project has not impacted habitat connectivity. The project is located outside critical wildlife corridors, ensuring unimpeded movement for terrestrial and aquatic species. No fragmentation of ecosystems has occurred, as natural water flow and vegetation continuity remain intact. Regular stakeholder consultations validate that the project does not disrupt ecological linkages, maintaining the integrity of local habitats.

2.4.2 Introduction of species

This section is not Applicable as this project does not introduce any species during this monitoring period.

2.4.3 Ecosystem conversion

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	No risk identified	The project is located in a non-sensitive area, avoiding forests, wetlands, and other critical ecosystems. No natural habitats have been cleared, degraded, or altered, as the project design prioritizes environmental sustainability. Continuous stakeholder consultations further validate that land use remains unchanged, with no transformation of ecosystems or biodiversity loss.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The project activity is a Hydro power project with a total installed capacity of 15 MW. The project is promoted by Nam Houg - 1 Hydro Power Co., Ltd which is a SPV of Simoung Group Co., Ltd. Simoung Group Co., Ltd. is the legal owner of Nam Houg-1 hydro power project. The Project activity is a new facility (Greenfield) and the electricity generated by the project will be exported to the electricity grid. The project will therefore displace an equivalent amount of electricity which would have otherwise been generated by fossil fuel dominant electricity grid.

The Project Proponent plans to avail the VCS benefits for the project. In the Pre- project scenario, the entire electricity delivered to the grid by the project activity, would have otherwise been generated by the operation of grid-connected power plant. The scenario prior to the start of implementation of the project activity is generation of electricity, dominated by thermal power plants, thus leading to GHG emissions. The baseline scenario is the same as the scenario prior to the start of implementation of the project activity. The project shall result in replacing anthropogenic emissions of greenhouse gases (GHG's) estimated to be approximately 31,453 tCO_{2e} per year, thereon displacing 87,280 MWh/year amount of electricity from the grid over the 7 years crediting period. The hydro project structure consists of release sluice, main powerhouse, and right-bank non overflow dam, etc.

Main Technical Parameters of the project

Turbine		
Item	Value	Unit
Number of units	3	Set
Rated capacity	5	MW
Maximum working head	22	m
Least working head	14.67	m
Rated head	21	m
Rated speed	250	rpm
Rated flow	28.7	m ³ /s
Turbine type	Bulb	
Lifetime	30	years
Generator		
Item	Value	Unit
Number of units	3	Set
Rated capacity	5	MW
Generator rated power factor	0.85	
Generator efficiency	97	%
Lifetime	30	years

3.2 Deviations

3.2.1 Methodology Deviations

No methodology deviation is applied during this monitoring period.

3.2.2 Project Description Deviations

There is no change in project design related to installed capacity, technology, specification, etc.

3.3 Grouped Projects

The project is not a grouped project. Thus, this section is not applicable.

3.4 Baseline Reassessment

Did the project undergo baseline reassessment during the monitoring period?

☐ Yes

☒ No

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO ₂ / MWh
Description	Operating Margin CO ₂ emission factor in year y
Source of data	Calculated
Value applied	0.4504
Justification of choice of data or description of measurement methods and procedures applied	Calculated as per “Tool to calculate the emission factor for an electricity system, version 07” as 3-year generation weighted average using data for the years 2018, 2019 & 2020
Purpose of data	Calculation of baseline emissions
Comments	This parameter is fixed ex-ante for the entire crediting period

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO ₂ / MWh
Description	Built Margin CO ₂ emission factor in year y
Source of data	Calculated
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	Calculated as per “Tool to calculate the emission factor for an electricity system, version 07”
Purpose of data	Calculation of baseline emissions
Comments	This parameter is fixed ex-ante for the entire crediting period

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO ₂ /MWh
Description	Combined margin CO ₂ emission factor for the project electricity system in year y
Source of data	Calculated
Value applied	0.4504
Justification of choice of data or description of measurement methods and procedures applied	The combined margin emissions factor is calculated as follows: $EF_{grid,CM,y} = EF_{grid,OM,y} * WOM + EF_{grid,BM,y} * WBM$ Where: $EF_{grid,BM,y} = \text{Build margin CO}_2 \text{ emission factor in year y (tCO}_2\text{/MWh)}$

	$EF_{grid,OM,y}$ = Operating margin CO₂ emission factor in year y (tCO₂/MWh) WOM = Weighting of operating margin emissions factor (%) = 100% (for simplified CM) WBM= Weighting of build margin emissions factor (%) = 0% (for simplified CM)
Purpose of data	Calculation of baseline emissions
Comments	This parameter is fixed ex-ante for the entire crediting period

Data / Parameter	EF_{Res}
Data unit	kgCO ₂ e/MWh
Description	Default emission factor for emissions from reservoirs
Source of data	ACM0002: Grid-connected electricity generation from renewable sources V.21.0
Value applied	90 kgCO ₂ e/MWh
Justification of choice of data or description of measurement methods and procedures applied	Not Applicable
Purpose of data	Calculation of Power Density
Comments	This parameter is fixed ex-ante for the entire crediting period

Data / Parameter	Cap_{BL}
Data unit	W
Description	Installed capacity of the hydro power plant before the implementation of the project activity. For new hydro power plants, this value is zero
Source of data	ACM0002: Grid-connected electricity generation from renewable sources V.21.0
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	Not Applicable
Purpose of data	Calculation of Power Density
Comments	This parameter is fixed ex-ante for the entire crediting period

Data / Parameter	ABL
Data unit	m ²
Description	Area of the single or multiple reservoirs measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m ²). For new reservoirs, this value is zero
Source of data	Project site
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	Since this is project involves construction of new reservoirs, it is considered as zero as per methodology.
Purpose of data	Calculation of project emissions
Comments	This parameter is fixed ex-ante for the entire crediting period

4.2 Data and Parameters Monitored

Data / Parameter	EG _{PJ,y}
Data unit	MWh/y
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y in MWh
Source of data	Electricity meter readings
Description of measurement methods and procedures to be applied	The difference of final value of export and import is used for monthly values of net electricity supplied to the grid by the project activity and same value will be considered for ER calculations.
Frequency of monitoring/recording	Monthly
Value monitored	72,978 MWh
Monitoring equipment	The electricity exported / supplied by the plant to pooling substation and further to substation. This meter also measures electricity imported by the plant from the grid. Energy meter with an accuracy: class 1.0 for kWh, class 2.0 for kvarh Energy meter make – DTSD341 Serial Number: 1. Main – 250318494, Backup – 250318495 2. Main – 250318493, Backup – 219014741

QA/QC procedures to be applied	The frequency of calibration of energy meters is once in 5 years. The monthly electricity supplied/exported by the project activity in the JMR report is cross checked with the monthly invoices of sale. In the absence or delay in the meter calibration appropriate Guidelines will be applied to confirm the conservativeness of metering. The metering arrangement, accuracy class of meters, calibration frequency is under control of EDL and PP does not have any control on it.
Purpose of the data	Calculation of baseline emissions
Calculation method	Net electricity supplied to the grid by the project plant in a given month = Export(kWh)– Import (kWh)
Comments	Data will be archived in paper & electronic form for two years after the end of crediting period or of the last issuance of VCU for this project activity, whichever occurs later.

Data / Parameter	TEGy
Data unit	MWh/year
Description	Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads, in year y
Source of data	Project Activity Site
Description of measurement methods and procedures to be applied	Electricity meters
Frequency of monitoring/recording	Continuous measurement and at least monthly recording
Value monitored	72,978
Monitoring equipment	Electricity meters
QA/QC procedures to be applied	NA
Purpose of the data	Calculation of <i>project</i> emissions
Calculation method	NA
Comments	NA

Data / Parameter	CapPJ
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Data unit	W
Description	Installed capacity of the hydro power plant after the implementation of the project activity
Source of data	Project Site
Description of measurement methods and procedures to be applied	Determine the installed capacity based on manufacturer's specifications or commissioning data or recognized standards
Frequency of monitoring/recording	Once at the beginning of each crediting period
Value monitored	15000000
Monitoring equipment	NA
QA/QC procedures to be applied	NA
Purpose of the data	Calculation of project emissions
Calculation method	NA
Comments	NA

Data / Parameter	APJ
Data unit	m ²
Description	Area of the single or multiple reservoirs measured in the surface of the water, after the implementation of the project activity, when the reservoir is full
Source of data	Project Site
Description of measurement methods and procedures to be applied	Measured from topographical surveys, maps, satellite pictures, etc
Frequency of monitoring/recording	Once at the beginning of each crediting period
Value monitored	1520000
Monitoring equipment	-
QA/QC procedures to be applied	-

Purpose of the data	Calculation of <i>project</i> emissions
Calculation method	-
Comments	-

4.3 Monitoring Plan

The monitoring plan is developed to establish suitable data collection method for measurement & collection of data and maintenance of records according to the monitoring methodology of AMS I.D Version 18.0. The monitoring plan is project specific for which, the project performance with all relevant criteria will be monitored. Proper training will be provided to concerned personnel for operation purpose.

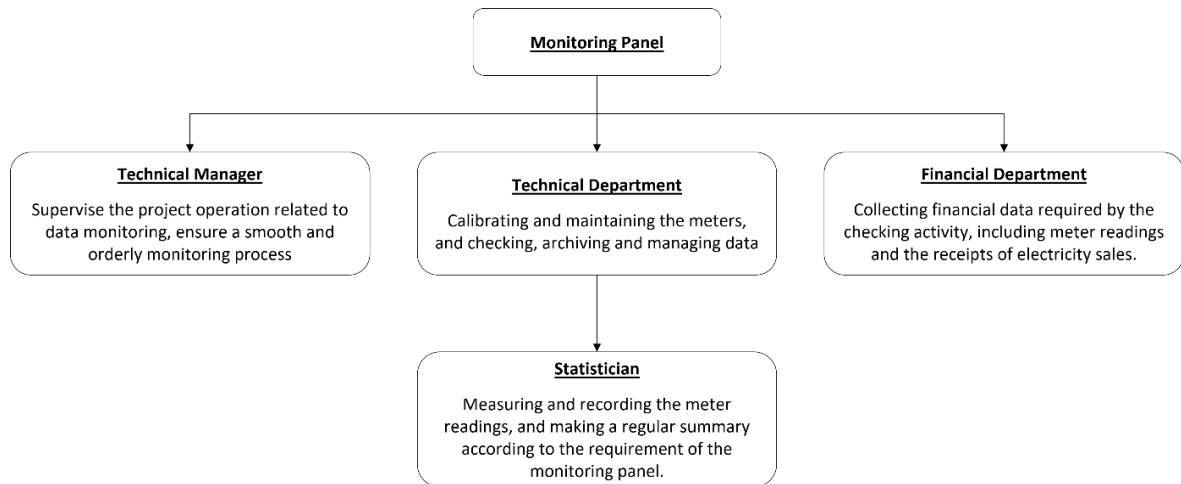
The purpose of monitoring is to calculate and monitor GHG emission reduction by the project activity. The monitoring plan, which is implemented by the project participant describes about the following aspects:

- Overall project management
- Monitoring Plan
- Emergency Preparedness
- Training on Monitoring & Archiving of Data and Internal Audit Procedures
- Quality Assurance and Quality Control (QA/QC) procedures

The project owner may seek support from the consultant to ensure quality of the collected data and monitoring procedures to be in line with the VCS requirements.

Overall project management team

The project owner organizes a separate team to be responsible for data collection, supervision and witness the whole process of data measuring and recording. The technical manager was appointed to take full responsibility for the overall monitoring of the project. The monitoring and measurement were carried out by designated technical manager. The technical department was responsible for carrying out the internal auditing and AQ/AC. All the values from generation records were cross checked with the invoices for consistency. The operation and management structure are shown below:



Structure of the monitoring team

Monitoring plan

The meter(s) were installed at the project site, to monitor the input/output electricity at the grid side. The meter(s) were installed in accordance with relevant national or international standard. Before the operation of the project, the metering equipment(s) were clarified and examined by the project owner and the power grid company according to the above regulation.

Personal Training:

The project employed qualified and experienced persons for plant operation. The training period shall be for three months, as this would be adequate and necessary to ensure proper imparting of the objective. The training course will be thoroughly and meticulously designed, highlighting the objectives, salient features, operational aspects and trouble shooting.

Procedure for handling of identified non-conformities

- The non-conformities are identified through routine inspections, monitoring systems, audits, or incident reports.
- The details of the non-conformity, including the date, time, location, and description of the event will be recorded.
- The plant operator will report the event immediately to the supervisor. Following, an initial assessment will be conducted to determine the severity and potential impact of the non-conformity on safety, operations, and compliance.
- This include stopping the process, isolating equipment, or implementing temporary fixes. Once the non-conformities are resolved, preventive actions will be implemented to avoid similar non-conformities in the future.
- This includes revising procedures, enhancing training programs, or upgrading equipment.
- The monitoring plan will be followed to continuously check for potential non-conformities and preventive measures will be reviewed periodically.

Emergency preparedness:

In case of any unforeseen event that is not covered under this monitoring plan, staff of the operation division will immediately inform the chief of the operation division. The chief of the operation division is then responsible to ensure that the cause for the unforeseen event is detected, the event is remedied and for the period of time in which the unforeseen event has occurred uncertainty in data gathered is limited as much as possible.

Data Collection:

The specific steps for data collection and reporting are listed below:

- a) During the crediting period, both the EDL and the project owner will record the values displayed by the main meter.
- b) Simultaneously to step a), the project owner will record the values displayed by the backup meters.
- c) The meters will be calibrated according to the relevant regulation and request of EDL
- d) The main meter's readings will be cross-checked with record document confirmed by EDL.
- e) The project owner and the EDL will record both output and input power readings from the main meter. These data will be used to calculate the amount of net electricity delivered to the grid.
- f) The project owner will be responsible of providing copies of record document confirmed by EDL to the DOE for verification.

If the reading of the main meter in a certain month is inaccurate and beyond the allowable error or the meter doesn't work normally, the grid-connected power generation shall be determined by following measures:

- a) Read the data of the backup meters.
- b) If the backup meter's data is not so accurate as to be accepted, or the practice is not standardized, the project owner and the grid corporation should jointly make a reasonable and conservative estimation method which can be supported by sufficient evidence and proved to be reasonable and conservative when verified by DOE.
- c) If the project owner and the EDL don't agree on an estimated method, arbitration will be conducted according the procedures set by the agreement to work out an estimation method.

Internal Auditing:

Project owner will conduct the internal auditing by cross checking the data from the invoices and data logbooks for ensuring the quality and consistency of data measured and monitored for the purpose of emission reduction calculations.

Quality Assurance and Quality Control (QA/QC) procedures**Calibration of Measuring Equipment's:**

The reliability of the monitoring system depends on the accuracy of the measuring instrument and the quality of the relevant equipment. Thus, the meters will be regularly calibrated as per the frequency specified by the manufacturer. To assist in future verifications, the project owner will preserve the calibration records, along with the data files of project monitoring. Error check routines will be established on site and at the point of data storage to detect data measuring as well as malfunctions.

In the case of malfunction of the meters, the meter supplier will provide technical support to engage the problem promptly and emission reductions during the corresponding period will be calculated conservatively.

Monitoring Report

Physical document such as the plant electrical wiring diagram will be gathered with this monitoring plan in a single place. In order to facilitate auditors' access to project documents, the project materials and monitoring results will be indexed. All paper-based information will be stored by the technical department of the project owner and all the material will have a copy for backup. All data, including calibration records, will be kept until 2 years after the end of the total crediting period.

During the crediting period, at the end of each year, the monitoring officer shall produce a monitoring report covering the past monitoring period. The report shall be transmitted to the General Manager who will check the data and issue a final monitoring report in the name of the project participants. Once the final report is issued, it will be submitted to the DOE for verification.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

Baseline emissions include only CO₂ emissions from electricity generation in power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,y} \quad (\text{Equation 1})$$

Where,

BE_y = Baseline emissions in year y (t CO₂/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)

$EF_{grid,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO₂/MWh)

If the project activity is the installation of a greenfield power plant, then:

$$EG_{PJ,y} = EG_{PJ,facility,y} \quad (\text{Equation 2})$$

Where,

$EG_{PJ,facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh)

Calculate the $EF_{grid,y}$

The emission factor should be calculated in a transparent and conservative manner according to the procedures prescribed in the “Tool to calculate the emission factor for an electricity system” (Version 07.0).

The data used for calculation are from an official source (where available) and publicly available. The calculation processes are as follows:

- STEP1: Identify the relevant electricity system;
- STEP2: Choose whether to include off-grid power plants in the project electricity system (optional);
- STEP3: Select a method to determine the operating margin (OM);
- STEP4: Calculate the operating margin emission factor according to the selected method;
- STEP5: Calculate the build margin (BM) emission factor;
- STEP6: Calculate the combined margin (CM) emissions factor.

STEP 1: Identify the relevant electricity system

As described in Section B.3., there are no transmission constraints between Lao and Thailand, the project electricity system is a regional grid consisting of Thailand Power Grid and Lao Power Grid. And as electricity imported from Malaysia, China and Vietnam Power Grid¹⁰, these three Power Grids are considered as connected electricity system.

STEP 2: Choose whether to include off-grid power plants in the project electricity system (optional)

According to “Tool to calculate the emission factor for an electricity system” (Version 07.0), there are two options to calculate the operating margin and build margin emission factor:

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.

Option I is chosen for emission factor calculation.

STEP 3: Select a method to determine the operating margin (OM)

According to “Tool to calculate the emission factor for an electricity system” (Version 07.0), there are four methods for calculating the $EF_{grid,OM,y}$:

- a) Simple OM, or
- b) Simple adjusted OM, or
- c) Dispatch data Analysis OM, or
- d) Average OM

The method (d), average OM, is selected.

$EF_{grid,OM-ave,y}$ is calculated using ex ante option: a 3-year generation-weighted average in 2020, 2019, 2018 without requirement to monitor and recalculate the emissions factor during the crediting period.

STEP 4: Calculate the operating margin emission factor according to the selected method

The average OM emission factor is calculated as the average emission rate of all power plants serving the grid, using the methodological guidance as described under Step 4 in the “Tool to calculate the emission factor for an electricity system” (Version 07.0) for the simple OM, but also including the low-cost / must-run power plants in all equations.

According to “Tool to calculate the emission factor for an electricity system” (Version 07.0), there are two options based on different data for calculating average OM:

Option A: Based on the net electricity generation and a CO₂ emission factor of each power unit;
Or

Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

For the project, the necessary data for Option A is not available, so Option B was used.

Under this option, the average OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, including low-cost/must-run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$$EF_{grid,OM-ave,y} = \frac{\sum_i (FC_{i,y} \times NCV_{i,y} \times EF_{CO_2,i,y})}{EG_y}$$

$EF_{grid,OM-ave,y}$ $FC_{i,y}$ = Average operating margin CO₂ emission factor in year y (tCO₂/MWh)

$FC_{i,y}$ = Amount of fossil fuel type i consumed in the project electricity system in year y (mass or volume unit)

$NCV_{i,y}$ = Net calorific value (energy content) of fossil fuel type i in year y (GJ/mass or volume unit)

$EF_{CO_2,i,y}$ = CO₂ emission factor of fossil fuel type i in year y (tCO₂/GJ)

EG_y = Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must-run power plants/units, in year y (MWh)

I = All fossil fuel types combusted in power sources in project electricity system in year y

y = The data available in the most recent 3 years

According to the “Tool to calculate the emission factor for an electricity system” (Version 07.0), electricity imports from the connected electricity systems $EG_{import,y}$ are included in the EG_y

The calculation value of average OM emission factor is 0.5542 tCO₂/MWh and the detailed calculating procedures please refer to ER calculation sheet.

Step 5. Calculate the build margin (BM) emission factor

To calculate the build margin (BM) emission factor, the data for determine the sample group of power units m about the most recently units in the electricity system is needed. However, as an international project system, it's difficult to obtain the information for all the units in both Lao and Thailand (power generation data, commissioning date, and the fuel consumption). The data requirements for the application for calculate the build margin (BM) emission factor cannot be met.

As the Simplified CM is adopted in the step 6, the weighting of build margin emissions factor is 0.

STEP 6: Calculate the combined margin (CM) emissions factor

According to "Tool to calculate the emission factor for an electricity system" (Version 07.0), the calculation of the combined margin (CM) emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

- a) Weighted average CM; or
- b) Simplified CM.

According to "Tool to calculate the emission factor for an electricity system" Version 07.0, the simplified CM can be used if:

- a) The project activity is located in: (i) a Least Developed Country (LDC); or in (ii) a country with less than 10 registered CDM projects at the starting date of validation; or (iii) a Small Island Developing States (SIDS); and
- b) The data requirements for the application of Step 5 above cannot be met.

The project located in Lao, which is a Least Developed Country, therefore the criteria (a) is met; As mentioned in step 5, the data requirements for the application for calculate the build margin (BM) emission factor is not available, therefore the criteria (b) is also met.

The Simplified CM method is calculated as follow:

$$EF_{grid,CM,y} = w_{OM} \times EF_{grid,OM,y} + w_{BM} \times EF_{grid,BM,y}$$

Where:

w_{OM} = Weighting of operating margin emission factor (%)

w_{BM} = Weighting of build margin emission factor (%).

The weighs w_{OM} and w_{BM} , for simplified CM by default, are $w_{OM}=1$ and $w_{BM}= 0$.

Hence calculation value of CM emission factor is 0.4504 tCO₂/MWh

Baseline Emission Reduction for the Monitoring period:

Year	Net Generation (MWh)	Emission factor (tCO ₂ /MWh)	Baseline Emission Reductions (tCO ₂ e)
01-September-2023 to 31-December-2023	24,107	0.45037	10,857
01-January-2024 to 31-December-2024	48,871	0.45037	22,010
Total	72,978		32,867

Electricity generation details:

Month	Export (kWh)	Import (kWh)	Net electricity (kWh)	Net electricity (MWh)
September 2023	6,487,161	595	6,486,566	6,486
October 2023	7,899,609	114	7,899,495	7,899
November 2023	6,334,805	49	6,334,756	6,334
December 2023	3,388,433	0	3,388,433	3,388
January 2024	1,848,916	1,789	1,847,127	1,847
February 2024	1,514,827	2,167	1,512,660	1,512
March 2024	1,167,805	4,757	1,163,048	1,163
April 2024	891,989	8,597	883,392	883
May 2024	2,328,187	9,192	2,318,995	2,318
June 2024	3,429,961	3,376	3,426,585	3,426
July 2024	6,453,860	70	6,453,790	6,453
August 2024	9,907,621	136	9,907,485	9,907
September 2024	7,687,608	3,482	7,684,126	7,684
October 2024	8,025,650	50	8,025,600	8,025
November 2024	3,450,528	45	3,450,483	3,450
December 2024	2,203,224	23	2,203,201	2,203

5.2 Project Emissions

The project activity is a hydro power project activity and as per the para 39 of the applied methodology the Emissions from water reservoirs of hydro power plants shall be determined considered following the procedure described in the most recent version of “ACM0002: Grid-connected electricity generation from renewable sources”. As per ACM0002: Grid-connected electricity generation from renewable sources V.21.0”, paragraph 35

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y} \quad (\text{Equation 1})$$

For this project activity, the emissions from water reservoirs of hydro power plants ($PE_{HP,y}$) are applicable. The power density (PD) of the project activity is calculated as follows:

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

Where:

PD = Power density of the project activity (W/m²)

Cap_{PJ} = Installed capacity of the hydro power plant after the implementation of the project activity (W) (ie, 15,000,000 W)

Cap_{BL} = Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero

A_{PJ} = Area of the single or multiple reservoirs measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (m²) (ie, as per the feasibility report it is 1520000 m²)

A_{BL} = Area of the single or multiple reservoirs measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m²). For new reservoirs, this value is zero

$$PD = (15,000,000-0)/(1,520,000-0) = 9.87 \text{ W/m}^2$$

As per ACM0002, version 21.0 methodology, If the power density of the project activity is greater than 4 W/m² and less than or equal to 10 W/m², $PE_{HP,y}$ is calculated

$$PE_{HP,y} = \frac{EF_{Res} \times TEG_y}{1000}$$

Where:

$PE_{HP,y}$ = Project emissions from water reservoirs (t CO_{2e}/yr)

EF_{Res} = Default emission factor for emissions from reservoirs of hydro power plants (kg CO_{2e}/MWh) -As per ACM0002, EF_{Res} table in page number 13, it is 90 kgCO_{2e}/MWh

TEG_y = Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads, in year y (MWh)

Project Emissions during the Monitoring period:

Year	Net Generation (MWh)	EF_{Res} (kgCO ₂ /MWh)	Project Emissions (tCO _{2e})
01-September-2023 to 31-December-2023	24,107	90	2,170

01-January-2024 to 31-December-2024	48,871	90	4,398
Total	72,978		6,568

5.3 Leakage Emissions

No leakage emissions are considered.

5.4 GHG Emission Reductions and Carbon Dioxide Removals

The ex-ante emission reductions (ER_y) for the project activity are calculated as follows

$$ER_y = BE_y - PE_y - LE_y$$

Where,

ER_y = Emission Reduction in year y (tCO_2)

BE_y = Baseline emission in year y (tCO_2)

PE_y = Project emissions in year y (tCO_2)

LE_y = Leakage Emissions in year y (tCO_2)

Vintage period	Baseline emissions (tCO_2e)	Project emissions (tCO_2e)	Leakage emissions (tCO_2e)	Reduction VCU's (tCO_2e)	Removal VCU's (tCO_2e)	Total VCU's (tCO_2e)
01-September-2023 to 31-December-2023	10,857	2,170	0	8,687	0	8,687
01-January-2024 to 31-December-2024	22,010	4,398	0	17,611	0	17,611
Total	32,867	6,568	0	26,298	0	26,298

Comparison of estimated emission reduction with actual emission reduction:

Parameters	Vintage Period		Total
Start date of monitoring period	01-September-2023	01-January-2024	01-September-2023
End date of Monitoring period	31-December-23	31-December-2024	31-December-2024
Total days of generation in the monitoring period	122	366	488

Estimated emission Reduction (tCO2e)	10,513	31,539	42,052
Actual Emission Reduction (tCO2e)	8,687	17,611	26,298
Difference in Emission reduction	-17%	-44%	-37%

The percentage difference between actual GHG emission reduction and estimated GHG emission reduction on yearly basis has been mentioned below in the table.

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
01-September-2023 to 31-December-2023	10,513	8,687	-17%	The seasonal variations in the amount of rainfall and temperature affect the monthly electricity generation from the hydro power plant. Hence, the actual electricity generated from the project is less than the estimated value.
01-January-2024 to 31-December-2024	31,539	17,611	-44%	The seasonal variations in the amount of rainfall and temperature affect the monthly electricity generation from the hydro power plant. Hence, the actual electricity generated from the project is less than the estimated value.
Total	42,052	26,298	-37%	The seasonal variations in the amount of rainfall and temperature affect the monthly electricity generation from the hydro power plant. Hence, the actual electricity generated from the project is less than the estimated value.

APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

Not Applicable.

Section	Information	Justification
Not Applicable	Not Applicable	Not Applicable